

Tuples and dictionaries

Python uses lists for keeping data in order. It also has other data types for storing information called “tuples” and “dictionaries”. Data types such as these, which hold lots of items, are called “containers”.

SEE ALSO

◀ 110–111 Types of data

◀ 128–129 Lists

Tuples

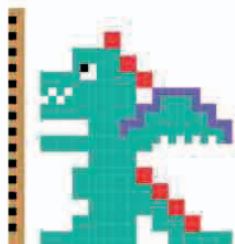
Tuples are a bit like lists, but the items inside them can't be changed. Once a tuple is set up it always stays the same.



Tuples are surrounded by brackets

```
>>> dragonA = ('Sam', 15, 1.70)
>>> dragonB = ('Fiona', 16, 1.68)
```

The items in a tuple are separated by commas



◁ What is a tuple?

A tuple contains items separated by commas and surrounded by brackets. Tuples are useful for collecting several bits of data together, such as a dragon's name, age, and height.

▷ Grabbing an item from a tuple

To get an item from a tuple, use its position in the tuple (its index). Tuples count from zero, just like lists and strings.

```
>>> dragonB[2]
1.68
```

This selects the item from position 2

```
>>> name, age, height = dragonA
>>> print(name, age, height)
Sam 15 1.7
```

The items that make up the tuple “dragonA” are displayed separately

◁ Splitting a tuple into variables

Assign three variables to the tuple “dragonA”—“name”, “age”, and “height”. Python splits the tuple into three items, putting one in each variable.

▷ Putting tuples in a list

Tuples can be put into a list because containers can go inside each other. Use this code to create a list of tuples.

```
>>> dragons = [dragonA, dragonB]
>>> print(dragons)
[('Sam', 15, 1.7), ('Fiona', 16, 1.68)]
```

Create a list of tuples called “dragons”

Lists go in square brackets

Each tuple is surrounded by round brackets inside the list's square brackets

Python displays all the items that are in the list, not just the names of the tuples